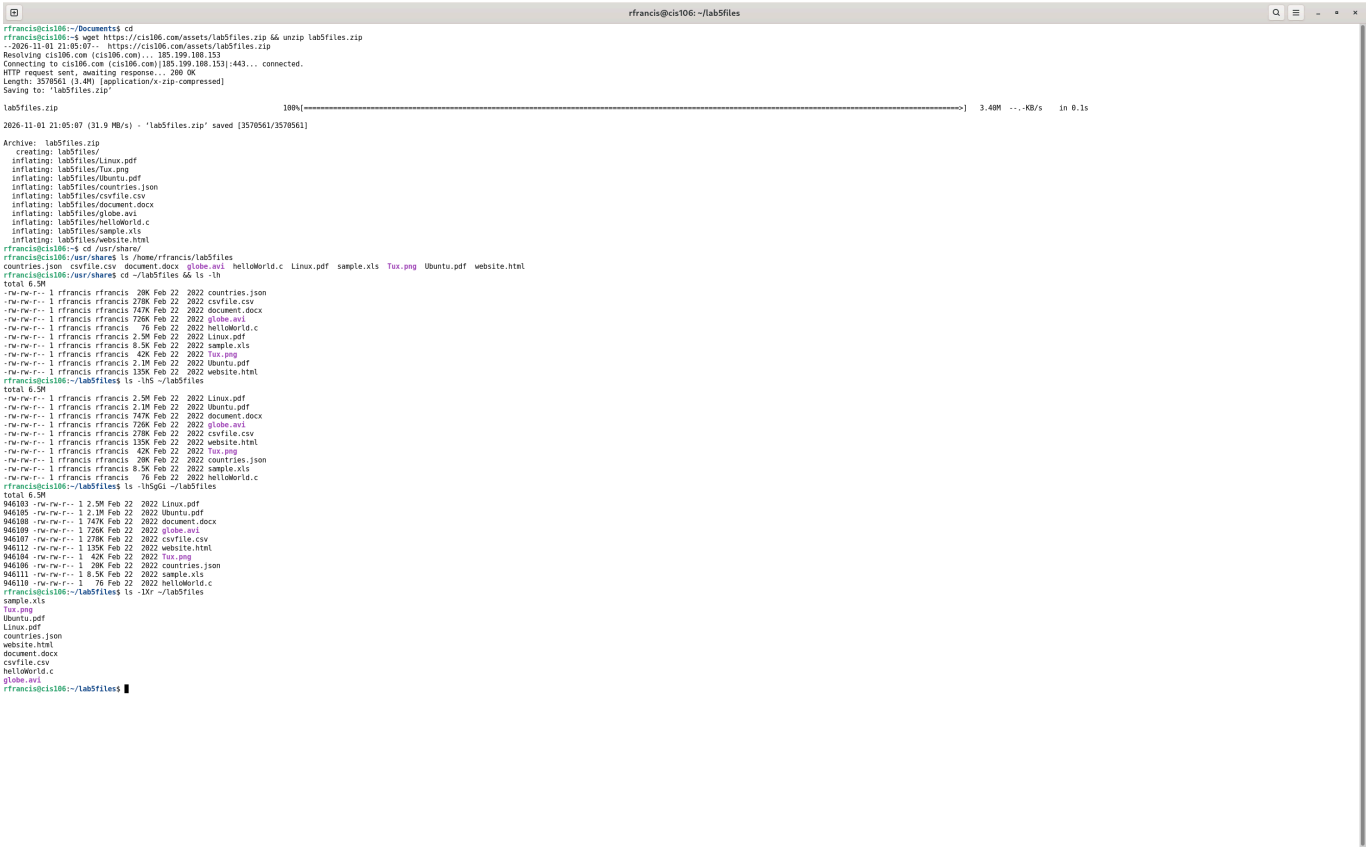
Lab 5: The Linux File System

Question 1

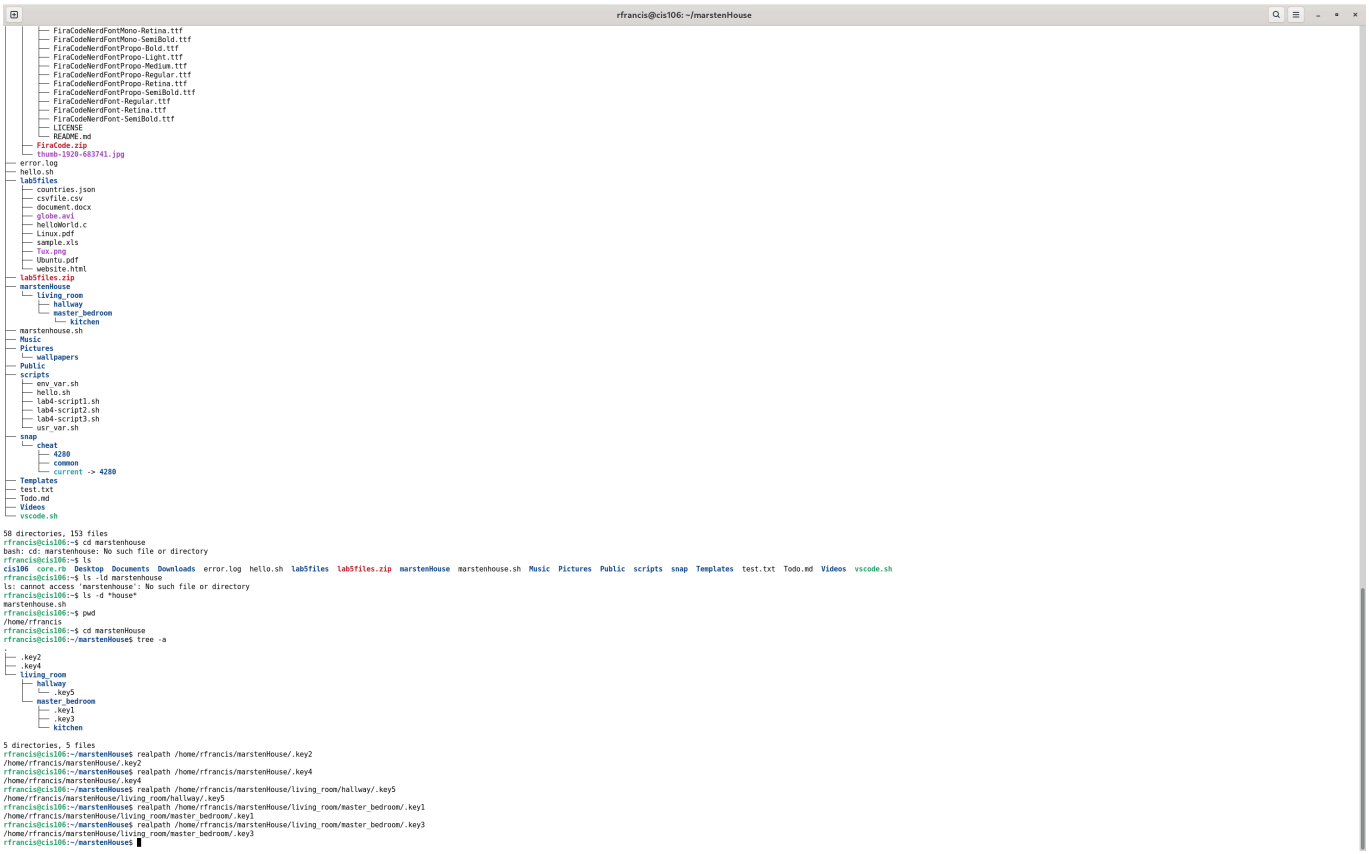




Question 2


rfrancis@cis106: ~/lab5files
rfrancis@cis106:~/Documents$ cd
rfrancis@cis106:~$ wget https://cis106.com/assets/lab5files.zip && unzip lab5files.zip
--2026-11-01 21:05:07-- https://cis106.com/assets/lab5files.zip
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3570561 (3.4M) [application/x-zip-compressed]
Saving to: 'lab5files.zip'
lab5files.zip 100%[==========>] 3.40M --.-KB/s in 0.1s
2026-11-01 21:05:07 (31.9 MB/s) - 'lab5files.zip' saved [3570561/3570561]
Archive: lab5files.zip
creating: lab5files/
inflating: lab5files/Linux.pdf
inflating: lab5files/Tux.png
inflating: lab5files/Ubuntu.pdf
inflating: lab5files/countries.json
inflating: lab5files/csvfile.csv
inflating: lab5files/document.docx
inflating: lab5files/globe.avi
inflating: lab5files/helloWorld.c
inflating: lab5files/sample.xls
inflating: lab5files/website.html
rfrancis@cis106:~$ cd /usr/share/
rfrancis@cis106:/usr/share$ ls /home/rfrancis/lab5files
countries.json csvfile.csv document.docx globe.avi helloWorld.c Linux.pdf sample.xls Tux.png Ubuntu.pdf website.html
rfrancis@cis106:/usr/share$ cd ~/lab5files && ls -lh
total 6.5M
-rw-rw-r-- 1 rfrancis rfrancis 20K Feb 22 2022 countries.json
-rw-rw-r-- 1 rfrancis rfrancis 278K Feb 22 2022 csvfile.csv
-rw-rw-r-- 1 rfrancis rfrancis 747K Feb 22 2022 document.docx
-rw-rw-r-- 1 rfrancis rfrancis 726K Feb 22 2022 globe.avi
-rw-rw-r-- 1 rfrancis rfrancis 76 Feb 22 2022 helloWorld.c
-rw-rw-r-- 1 rfrancis rfrancis 2.5M Feb 22 2022 Linux.pdf
-rw-rw-r-- 1 rfrancis rfrancis 8.5K Feb 22 2022 sample.xls
-rw-rw-r-- 1 rfrancis rfrancis 42K Feb 22 2022 Tux.png
-rw-rw-r-- 1 rfrancis rfrancis 2.1M Feb 22 2022 Ubuntu.pdf
-rw-rw-r-- 1 rfrancis rfrancis 135K Feb 22 2022 website.html
rfrancis@cis106:~/lab5files$ ls -lhS ~/lab5files
total 6.5M
-rw-rw-r-- 1 rfrancis rfrancis 2.5M Feb 22 2022 Linux.pdf
-rw-rw-r-- 1 rfrancis rfrancis 2.1M Feb 22 2022 Ubuntu.pdf
-rw-rw-r-- 1 rfrancis rfrancis 747K Feb 22 2022 document.docx
-rw-rw-r-- 1 rfrancis rfrancis 726K Feb 22 2022 globe.avi
-rw-rw-r-- 1 rfrancis rfrancis 278K Feb 22 2022 csvfile.csv
-rw-rw-r-- 1 rfrancis rfrancis 135K Feb 22 2022 website.html
-rw-rw-r-- 1 rfrancis rfrancis 42K Feb 22 2022 Tux.png
-rw-rw-r-- 1 rfrancis rfrancis 20K Feb 22 2022 countries.json
-rw-rw-r-- 1 rfrancis rfrancis 8.5K Feb 22 2022 sample.xls
-rw-rw-r-- 1 rfrancis rfrancis 76 Feb 22 2022 helloWorld.c
rfrancis@cis106:~/lab5files$ ls -lhSgGi ~/lab5files
total 6.5M
946103 -rw-rw-r-- 1 2.5M Feb 22 2022 Linux.pdf
946105 -rw-rw-r-- 1 2.1M Feb 22 2022 Ubuntu.pdf
946108 -rw-rw-r-- 1 747K Feb 22 2022 document.docx
946109 -rw-rw-r-- 1 726K Feb 22 2022 globe.avi
946107 -rw-rw-r-- 1 278K Feb 22 2022 csvfile.csv
946112 -rw-rw-r-- 1 135K Feb 22 2022 website.html
946104 -rw-rw-r-- 1 42K Feb 22 2022 Tux.png
946106 -rw-rw-r-- 1 20K Feb 22 2022 countries.json
946111 -rw-rw-r-- 1 8.5K Feb 22 2022 sample.xls
946110 -rw-rw-r-- 1 76 Feb 22 2022 helloWorld.c
rfrancis@cis106:~/lab5files$ ls -1Xr ~/lab5files
sample.xls
Tux.png
Ubuntu.pdf
Linux.pdf
countries.json
website.html
document.docx
csvfile.csv
helloWorld.c
globe.avi
rfrancis@cis106:~/lab5files$

Question 3


rfrancis@cis106: ~/marstenHouse
│ │ ├── FiraCodeNerdFontMono-Retina.ttf
│ │ ├── FiraCodeNerdFontMono-SemiBold.ttf
│ │ ├── FiraCodeNerdFontPropo-Bold.ttf
│ │ ├── FiraCodeNerdFontPropo-Light.ttf
│ │ ├── FiraCodeNerdFontPropo-Medium.ttf
│ │ ├── FiraCodeNerdFontPropo-Regular.ttf
│ │ ├── FiraCodeNerdFontPropo-Retina.ttf
│ │ ├── FiraCodeNerdFontPropo-SemiBold.ttf
│ │ ├── FiraCodeNerdFont-Regular.ttf
│ │ ├── FiraCodeNerdFont-Retina.ttf
│ │ ├── FiraCodeNerdFont-SemiBold.ttf
│ │ ├── LICENSE
│ │ └── README.md
│ ├── FiraCode.zip
│ └── thumb-1920-683741.jpg
├── error.log
├── hello.sh
├── lab5files
│ ├── countries.json
│ ├── csvfile.csv
│ ├── document.docx
│ ├── globe.avi
│ ├── helloWorld.c
│ ├── Linux.pdf
│ ├── sample.xls
│ ├── Tux.png
│ ├── Ubuntu.pdf
│ └── website.html
├── lab5files.zip
├── marstenHouse
│ └── living_room
│ ├── hallway
│ └── master_bedroom
│ └── kitchen
├── marstenhouse.sh
├── Music
├── Pictures
│ └── wallpapers
├── Public
├── scripts
│ ├── env_var.sh
│ ├── hello.sh
│ ├── lab4-script1.sh
│ ├── lab4-script2.sh
│ ├── lab4-script3.sh
│ └── usr_var.sh
├── snap
│ └── cheat
│ ├── 4280
│ ├── common
│ └── current -> 4280
├── Templates
├── test.txt
├── Todo.md
├── Videos
└── vscode.sh
58 directories, 153 files
rfrancis@cis106:~$ cd marstenhouse
bash: cd: marstenhouse: No such file or directory
rfrancis@cis106:~$ ls
cis106 core.rb Desktop Documents Downloads error.log hello.sh lab5files lab5files.zip marstenHouse marstenhouse.sh Music Pictures Public scripts snap Templates test.txt Todo.md Videos vscode.sh
rfrancis@cis106:~$ ls -ld marstenhouse
ls: cannot access 'marstenhouse': No such file or directory
rfrancis@cis106:~$ ls -d *house*
marstenhouse.sh
rfrancis@cis106:~$ pwd
/home/rfrancis
rfrancis@cis106:~$ cd marstenHouse
rfrancis@cis106:~/marstenHouse$ tree -a
.
├── .key2
├── .key4
└── living_room
├── hallway
│ └── .key5
└── master_bedroom
├── .key1
├── .key3
└── kitchen
5 directories, 5 files
rfrancis@cis106:~/marstenHouse$ realpath /home/rfrancis/marstenHouse/.key2
/home/rfrancis/marstenHouse/.key2
rfrancis@cis106:~/marstenHouse$ realpath /home/rfrancis/marstenHouse/.key4
/home/rfrancis/marstenHouse/.key4
rfrancis@cis106:~/marstenHouse$ realpath /home/rfrancis/marstenHouse/living_room/hallway/.key5
/home/rfrancis/marstenHouse/living_room/hallway/.key5
rfrancis@cis106:~/marstenHouse$ realpath /home/rfrancis/marstenHouse/living_room/master_bedroom/.key1
/home/rfrancis/marstenHouse/living_room/master_bedroom/.key1
rfrancis@cis106:~/marstenHouse$ realpath /home/rfrancis/marstenHouse/living_room/master_bedroom/.key3
/home/rfrancis/marstenHouse/living_room/master_bedroom/.key3
rfrancis@cis106:~/marstenHouse$